

Bangladesh University of Engineering and Technology

CSE208: Data Structure and Algorithms Sessional II

Online 3

Section: B1 + B2

Time: 35 Minutes

The Stable Network

You are a network engineer for a high-security cluster of N servers. You have a list of M potential direct connections (cables) you can install between specific pairs of servers. Each connection has a specific **latency** value.

For the system to be stable, the network must be **connected** (any server can reach any other server). However, a large variance in latency causes synchronization bugs. Your goal is to select a subset of connections to connect all servers such that the **difference between the highest latency connection and the lowest latency connection** in your subset is minimized.

Input Format

- First line: Two integers N and M (number of servers, number of potential connections).
- Next M lines: Three integers u , v , w (A connection between server u and server v with latency w).

Output Format

Print a single integer: the minimum possible difference between the max and min latency in a valid connected network.

If it is impossible to connect all servers, print -1.

Constraints

- $2 \leq N \leq 200$
- $1 \leq M \leq 5,000$
- $0 \leq w \leq 10^9$

Samples

Sample 1

Input	Output
4 6 1 2 1 2 3 10 3 4 20 1 4 15 1 3 12 2 4 11	2

Explanation: Using edges with weights 10, 11, 12: (2-3), (2-4), (1-3). Difference = $|12-10| = 2$.

Sample 2

Input	Output
5 7 1 2 1 2 3 4 3 4 6 4 5 9 1 5 10 1 3 5	4

2 4 7	
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Sample 3

Input	Output
5 6 1 2 5 2 3 6 1 3 7 4 5 10 3 4 100 2 4 200	94